

# STUDENT PERFORMANCE ANALYTICS PROJECT REPORT

*A Data Analytics Portfolio Case Study*

Tools Used: Microsoft Excel | Power BI

Datasets: 100-Student Score Dataset | 300-Student Study-Habit Dataset

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## 1. Executive Summary

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This report presents a comprehensive analysis of student academic performance, built from a 100-student score dataset (Math, Reading, and Writing scores with demographic attributes) visualized in a Power BI dashboard, alongside a complementary 300-student study-habit dataset examining how study hours, sleep, screen time, and assignment completion relate to a classified performance level (Low, Medium, High). Together, the two dashboards move from “what scores look like” to “what behaviors are associated with stronger outcomes.”

Across the 100-student cohort, the average overall score is 61.18%, with Mathematics the strongest subject (62.51) and Reading the weakest (60.49). Performance is fairly even across gender (Female 61.26 vs Male 61.08) and ethnicity groups (Group A leads at 63.03; Group C trails at 60.04), but test preparation stands out as the single largest differentiator in the dataset — students who completed a preparation course scored 66.34 on average versus 57.59 for those who did not, an 8.75-point gap that is larger than any demographic gap observed.

The companion 300-student study-habit dashboard reinforces this behavioral pattern at a broader scale: students classified as High performers average 7.5 study hours versus 2.7 for Low performers, and Low performers report over four times the screen time of High performers (6.49 vs 2.24 hours). Assignment completion follows the same gradient, with High-performance students completing far more assignments (937) than Medium (607) or Low (260) performers.

The core opportunity emerging from both dashboards is behavioral, not demographic: structured preparation, consistent study time, assignment completion, and controlled screen time are associated with materially better outcomes, and these are levers that an institution can actually influence — unlike gender or ethnicity. The recommendations in this report focus on scaling test-preparation access, monitoring assignment completion as an early-warning signal, and supporting the lowest-scoring student segment identified in the dashboard.

## 2. Academic Problem Statement

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Educational institutions and academic coordinators need a clear, evidence-based view of how students are performing and which factors are associated with stronger or weaker outcomes. Without a consolidated view, performance gaps across preparation status, parental education background, or study habits can go unnoticed until they show up in final results.

The Student Performance Dashboard and the Student Performance Classification – EDA Dashboard address this by helping stakeholders:

- Monitor overall academic performance across a student cohort at a glance.
- Track key score metrics (Math, Reading, Writing, overall percentage) as KPIs.
- Compare performance across gender, ethnicity, and parental education background.
- Identify the highest- and lowest-scoring students for targeted support.
- Quantify the relationship between test preparation and outcomes.
- Understand how study hours, sleep, screen time, and assignment completion relate to a classified performance level.
- Support data-driven decisions on where to focus academic support resources.

## 3. Project Objectives

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- Analyze overall academic performance across the 100-student dataset.

- Monitor average scores in Mathematics, Reading, and Writing, and an overall score percentage.
- Compare performance across gender, ethnicity, and parental education groups.
- Identify the students with the highest and lowest overall scores.
- Measure the impact of test preparation completion on scores.
- Examine how study hours, sleep hours, and screen time relate to performance level in the companion dataset.
- Provide interactive filtering (by ethnicity, gender, parental education, test preparation) for deeper self-service analysis.
- Translate the findings into practical, actionable recommendations for academic stakeholders.

## 4. Dataset Overview

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### 4.1 Core Score Dataset (Excel + Power BI Dashboard 1)

**Total records:** 100 students

**Number of columns:** 8 (Student ID, Gender, Ethnicity, Parental Education, Test Preparation, Math Score, Reading Score, Writing Score)

**Granularity:** One row per individual student

Key fields:

- Student ID — unique identifier (S001–S100), used to drill into individual performers.
- Gender — categorical (Male / Female), 47 male and 53 female students.
- Ethnicity — five categorical groups (Group A–E), used for demographic comparison.
- Parental Education — six categorical levels from “Some High School” to “Master’s Degree.”
- Test Preparation — whether the student completed a preparation course (“Completed” / “None”); 41 completed, 59 did not.
- Math Score, Reading Score, Writing Score — numeric scores (0–100), the core performance metrics.

This structure supports comparison of academic outcomes across demographic and preparation-related factors, and calculation of an overall score percentage (the average of the three subject scores) used throughout the dashboard.

### 4.2 Companion Study-Habit Dataset (Power BI Dashboard 2 – EDA)

**Total records:** 300 students (a separate, larger cohort from the 100-student score dataset)

**Key fields:** Study Hours, Sleep Hours, Screen Time, Assignments Completed, Practice Score, and a derived Performance Level (Low / Medium / High)

This dashboard is an exploratory data analysis (EDA) view built to classify students into performance bands based on behavioral and lifestyle variables rather than subject scores. It is presented as a complementary, not identical, dataset — the student counts and variables differ from the 100-student core dataset, and the two should be read as two related but distinct analyses within the same portfolio project.

## 5. Data Preparation

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The following preparation steps were carried out ahead of building the Power BI visualizations:

- Reviewed the Excel dataset structure, column headers, and data types (text categories vs. numeric scores).

- Verified that Math, Reading, and Writing scores were stored as numeric values suitable for aggregation.
- Checked categorical fields (Gender, Ethnicity, Parental Education, Test Preparation) for consistent labeling before using them as dashboard filters/slicers.
- Reviewed the Student ID field to confirm each record represented a unique student.
- Derived an overall “Score Percentage” measure by averaging the three subject scores per student, used as the primary performance metric in the dashboard.
- Prepared the cleaned dataset as the data source for the Power BI report, enabling the KPI cards, bar charts, and slicers shown in the dashboard.
- For the companion EDA dashboard, the Performance Level classification (Low / Medium / High) was treated as a prepared, categorical output field used to segment the study-habit variables.

## 6. Key Performance Indicators (KPIs)

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The core dashboard surfaces five headline KPIs summarizing the 100-student cohort:

| KPI                   | Value | What it Measures / Why it Matters                                                                       |
|-----------------------|-------|---------------------------------------------------------------------------------------------------------|
| Student Count         | 100   | Size of the analyzed cohort; the denominator for every average in the dashboard.                        |
| Avg. Reading Score    | 60.49 | Average reading ability across the cohort; the lowest of the three subjects.                            |
| Avg. Writing Score    | 60.53 | Average writing ability; nearly identical to Reading, suggesting linked literacy skills.                |
| Avg. Maths Score      | 62.51 | Average numeracy ability; the strongest subject area for this cohort.                                   |
| Avg. Score Percentage | 61.18 | Overall performance metric (mean of the three subjects); the single best summary of cohort performance. |

Companion EDA dashboard KPIs (300-student behavioral cohort):

| KPI                    | Value | What it Measures / Why it Matters                                                       |
|------------------------|-------|-----------------------------------------------------------------------------------------|
| Count of Students      | 300   | Size of the behavioral/study-habit cohort (separate from the 100-student score cohort). |
| Average Study Hours    | 5.06  | Typical daily/weekly study time; a key input to the performance classification.         |
| Average Sleep Hours    | 7.59  | Typical sleep duration; a wellbeing indicator alongside performance.                    |
| Average Screen Time    | 4.32  | Average non-study screen exposure; inversely associated with performance level.         |
| Average Practice Score | 68.21 | Average score on practice assessments used to help derive performance level.            |

## 7. Score Analysis (Subject-Level Performance)

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Across the 100-student dataset, subject-level performance is close but not identical:

- Mathematics is the strongest subject at 62.51, ahead of Writing (60.53) and Reading (60.49).
- Reading and Writing scores sit almost 2 points below Math, suggesting numeracy is a relative strength for this cohort compared to literacy-based subjects.
- Correlation between subjects is weak (Math–Reading 0.13, Math–Writing 0.09, Reading–Writing 0.07), meaning a student’s performance in one subject is not a strong predictor of performance in another — support strategies should be subject-specific rather than assuming a single “strong” or “weak” student profile.
- The overall score distribution is concentrated in the middle: 50 of 100 students fall in the 60–70% band, with only 12 students above 70% and none above 80%, indicating no standout top tier and room to lift the broad middle of the cohort.

Business (academic) significance: because subject scores don’t move together, a single “overall score” KPI can mask subject-specific weaknesses — the dashboard’s per-subject KPIs are necessary, not just the blended percentage.

## 8. Performance by Demographic Group

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### 8.1 By Ethnicity

| Ethnicity Group | Students (n) | Avg. Score % |
|-----------------|--------------|--------------|
| Group A         | 26           | 63.03        |
| Group D         | 17           | 61.63        |
| Group B         | 14           | 60.60        |
| Group E         | 19           | 60.11        |
| Group C         | 24           | 60.04        |

Group A is both the largest (26 students) and the highest-performing group (63.03), while Group C, similarly sized (24 students), is the lowest performer (60.04). The spread across all five groups is under 3 points, so ethnicity is a comparatively minor differentiator relative to test preparation (see Section 10).

### 8.2 By Gender

Female students (n=53) average 61.26% versus 61.08% for male students (n=47) — a negligible 0.18-point difference overall. The dashboard does show subject-level nuance: females lead in Math (63.58 vs 61.30) and Reading (60.83 vs 60.11), while males lead in Writing (61.83 vs 59.38). Gender is not a meaningful driver of overall performance in this dataset, but the subject-level split may be worth monitoring if it persists in future cohorts.

## 9. Top and Bottom Performing Students

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Identifying individual students at the extremes helps target recognition and intervention:

| Rank  | Student ID | Avg. Score % | Ethnicity | Parental Education | Test Prep |
|-------|------------|--------------|-----------|--------------------|-----------|
| Top 1 | S011       | 78.00        | Group C   | High School        | Completed |
| Top 2 | S070       | 77.67        | Group D   | Master's Degree    | Completed |

| Rank  | Student ID | Avg. Score % | Ethnicity | Parental Education | Test Prep |
|-------|------------|--------------|-----------|--------------------|-----------|
| Top 3 | S044       | 76.00        | Group A   | Bachelor's Degree  | Completed |
| Top 4 | S069       | 75.33        | Group A   | Bachelor's Degree  | Completed |
| Top 5 | S098       | 75.00        | Group A   | Associate's Degree | Completed |

A clear pattern: all five top performers completed test preparation, regardless of ethnicity or parental education background — reinforcing that preparation, not demographic background, is the common thread among high achievers.

At the other end, the lowest-scoring students (e.g., S089 and S057 at 43.67%, S005 at 44.67%, S061 at 47.67%, S058 at 48.67%) share the opposite pattern: none of them completed test preparation. These students are strong candidates for targeted academic support.

## 10. Parental Education Background Analysis

| Parental Education | Students (n) | Avg. Score % |
|--------------------|--------------|--------------|
| Some College       | 13           | 63.31        |
| Bachelor's Degree  | 17           | 63.12        |
| High School        | 20           | 61.67        |
| Some High School   | 14           | 60.74        |
| Associate's Degree | 23           | 59.51        |
| Master's Degree    | 13           | 59.18        |

Notably, the relationship is not linear: students whose parents completed “Some College” or a Bachelor’s Degree score highest (63.31 and 63.12), while the Master’s Degree group scores lowest (59.18) despite representing the highest formal education level. With only 13 students in that group, this likely reflects small-sample variation rather than a genuine negative effect of higher parental education, and should not be treated as a reliable trend without a larger sample.

## 11. Test Preparation Impact Analysis

This is the most decisive factor identified anywhere in the dashboard. Students who completed test preparation consistently and substantially outperform those who did not, in every single subject:

| Subject     | Completed Prep (Avg) | No Prep (Avg) | Gap   |
|-------------|----------------------|---------------|-------|
| Mathematics | 68.34                | 58.46         | +9.88 |
| Reading     | 64.61                | 57.63         | +6.98 |
| Writing     | 66.07                | 56.68         | +9.40 |
| Overall     | 66.34                | 57.59         | +8.75 |

With 41 of 100 students having completed preparation and 59 not having done so, more than half the cohort has not accessed a resource associated with an 8–10-point score improvement across every subject — this is the single highest-leverage opportunity in the dataset.

## 12. Score Distribution Analysis

| Score Band | Number of Students |
|------------|--------------------|
| Below 50%  | 10                 |
| 50% – 60%  | 28                 |
| 60% – 70%  | 50                 |
| 70% – 80%  | 12                 |
| 80%+       | 0                  |

Half the cohort (50 students) sits in the 60–70% band, and no student reaches 80% or above. This indicates a performance ceiling in the current dataset and a large “middle group” that preparation-style interventions could realistically move into the 70%+ band, based on the gap sizes observed in Section 11. Note: as this is a single-snapshot dataset rather than a time series, no monthly or seasonal trend is available or claimed.

## 13. Study-Habit and Performance-Level Analysis (Companion EDA Dashboard)

The 300-student EDA dashboard classifies students into Low, Medium, and High performance levels and examines four behavioral variables against that classification.

| Performance Level | Avg. Study Hours | Avg. Sleep Hours | Avg. Screen Time | Assignments Completed |
|-------------------|------------------|------------------|------------------|-----------------------|
| High              | 7.5              | 7.7              | 2.24             | 937                   |
| Medium            | 5.0              | 7.7              | 4.25             | 607                   |
| Low               | 2.7              | 7.3              | 6.49             | 260                   |

Three clear patterns emerge:

- Study hours scale directly with performance level — High performers study nearly 3x as many hours as Low performers (7.5 vs 2.7).
- Screen time scales inversely — Low performers report roughly 2.9x the screen time of High performers (6.49 vs 2.24 hours).
- Assignment completion shows the widest gap of all: High performers complete 937 assignments versus 260 for Low performers, a more than 3.5x difference — making it a strong candidate as an early-warning indicator.
- Sleep hours are comparatively flat across all three levels (7.3–7.7), suggesting sleep is not a strong differentiator in this dataset, unlike study time, screen time, and assignment completion.

Read together with the core dashboard, both datasets point to the same conclusion from different angles: measurable, coachable behaviors (test preparation, study time, assignment completion, screen time discipline) are associated with stronger outcomes far more than fixed demographic traits.

## 14. Power BI Dashboard Overview

### 14.1 Student Performance Dashboard (Core)

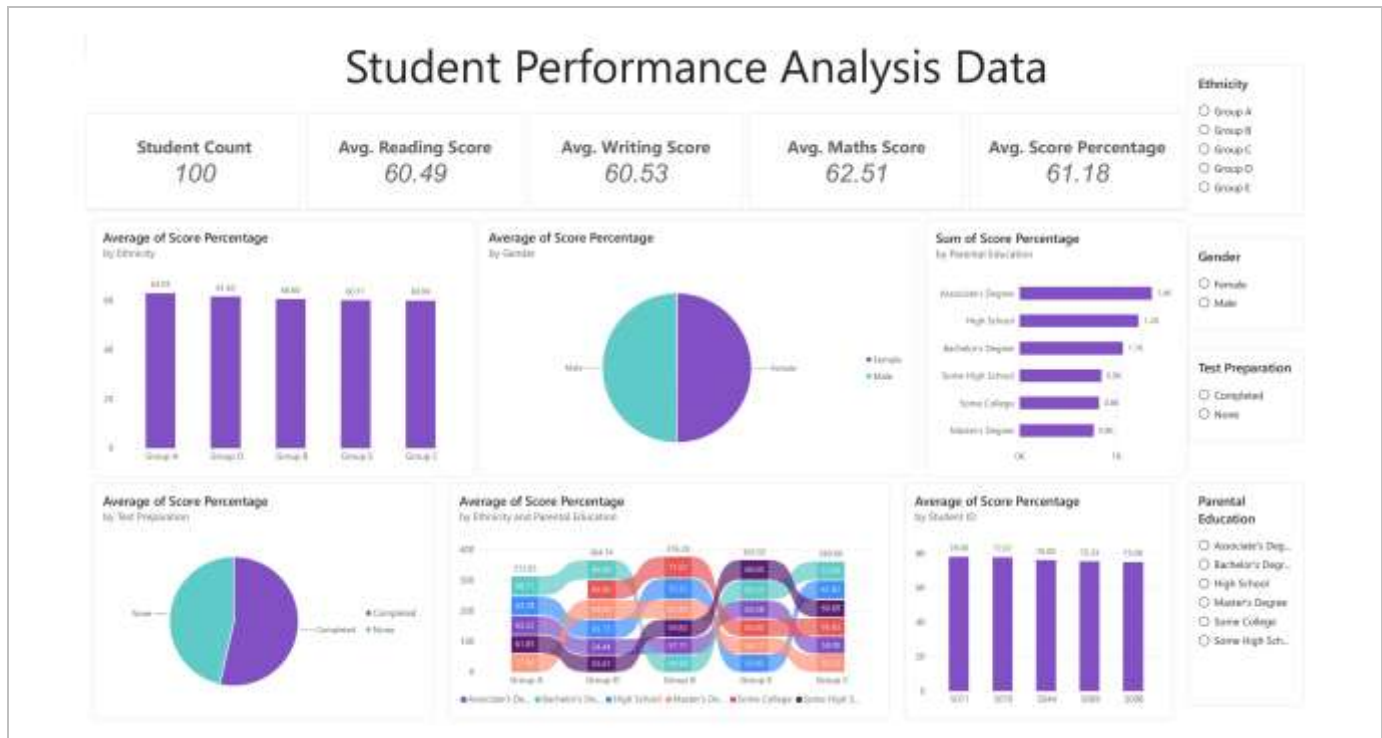


Figure 1: Student Performance Dashboard — 100-student core cohort (KPI cards, ethnicity, gender, parental education, test preparation, and top-student views).

This dashboard includes:

- **KPI Cards** — Student Count, Avg. Reading/Writing/Maths Score, Avg. Score Percentage. Answers: “How is the cohort performing right now, at a glance?”
- **Average Score Percentage by Ethnicity** (bar chart) — Answers: “Do outcomes differ by ethnic group?” Lets a coordinator quickly spot the highest/lowest-performing group (Group A vs Group C).
- **Average Score Percentage by Gender** (donut chart) — Answers: “Is there a gender performance gap?” Confirms the gap here is minimal.
- **Sum of Score Percentage by Parental Education** (bar chart) — Answers: “Does family educational background relate to performance?” Ranks education levels, though as a sum rather than average it is more useful for understanding group size-weighted contribution.
- **Average Score Percentage by Test Preparation** (donut chart) — Answers: “Does completing test prep matter?” The dashboard's clearest and most actionable signal.
- **Average Score Percentage by Ethnicity and Parental Education** (combo chart) — Answers: “Do ethnicity and parental education interact?” Lets stakeholders drill into any specific ethnicity/education combination.
- **Average Score Percentage by Student ID** (top-N bar chart) — Answers: “Who are the highest achievers?” Directly surfaces the top 5 students for recognition.

- Interactive Slicers — Ethnicity, Gender, Parental Education, Test Preparation. Let any stakeholder filter the entire dashboard to a specific student segment (e.g., “show me only females with no test preparation”) for self-service analysis.

## 14.2 Student Performance Classification – EDA Dashboard (Companion)

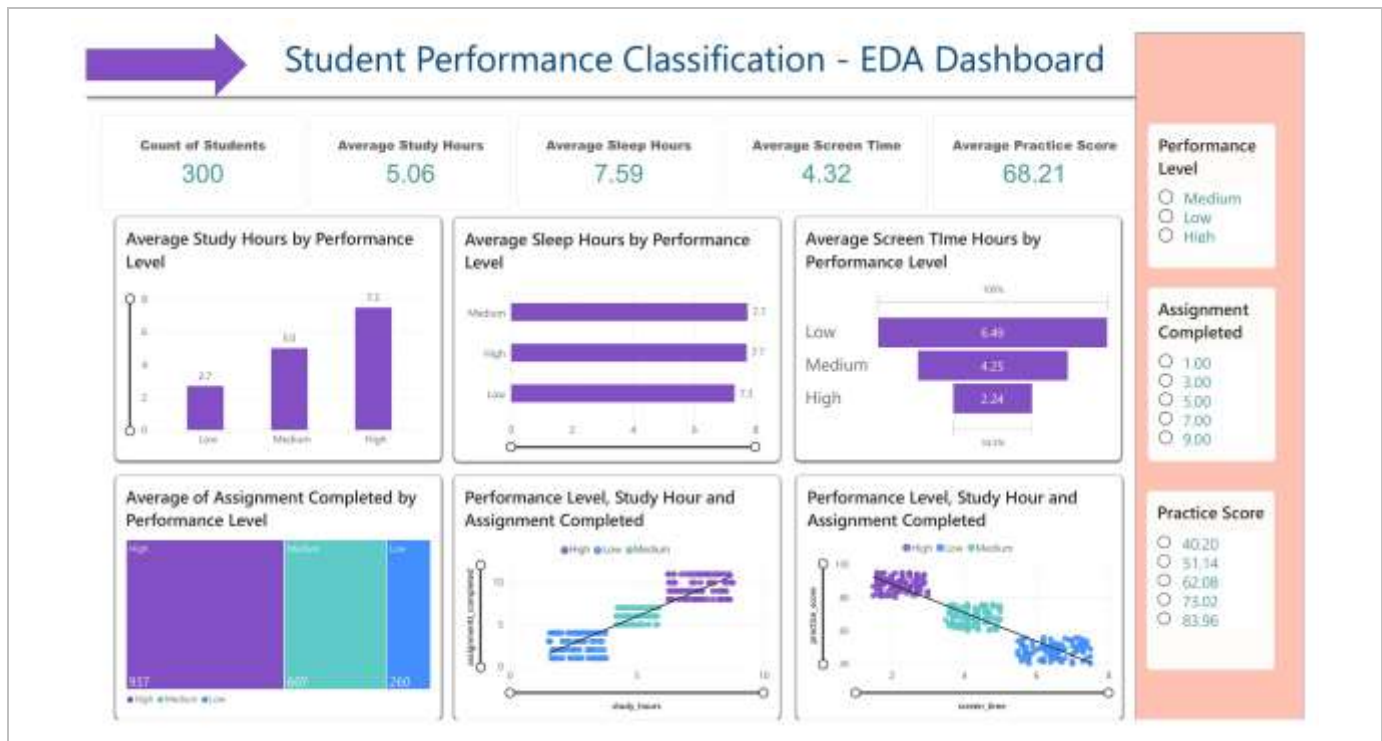


Figure 2: Student Performance Classification – EDA Dashboard — 300-student behavioral cohort (study hours, sleep, screen time, assignments completed, and practice score by performance level).

- KPI Cards — Count of Students, Average Study Hours, Sleep Hours, Screen Time, Practice Score. Establishes the behavioral baseline for the 300-student cohort.
- Average Study Hours by Performance Level (bar chart) — Answers: “How much more do top performers study?”
- Average Screen Time by Performance Level (donut/bar) — Answers: “Is screen time holding lower performers back?”
- Average Sleep Hours by Performance Level (bar chart) — Answers: “Does sleep differ by performance level?” (Here, the answer is: not much.)
- Performance Level, Study Hours and Assignments Completed (scatter plot) — Answers: “How do study hours and assignment completion jointly relate to performance level?” Useful for spotting students who study but don’t complete assignments, or vice versa.
- Average Assignments Completed by Performance Level (bar chart) — Answers: “Is assignment completion a leading indicator of performance?”
- Performance Level, Screen Time and Practice Score (scatter plot) — Answers: “Does screen time correspond to lower practice scores?” Helps visually confirm the inverse relationship.

## 15. Key Insights

**Insight:** Test preparation is the strongest performance driver in the dataset.

**Business Impact:** Completing preparation is associated with an 8.75-point overall score gap — larger than any gender, ethnicity, or parental-education gap observed.

**Recommended Action:** Prioritize expanding access to test preparation resources, especially for the 59% of students who have not completed one.

**Insight: Every one of the top 5 students completed test preparation.**

**Business Impact:** High achievement in this dataset is consistently linked to preparation rather than demographic background.

**Recommended Action:** Use top performers as case studies to promote preparation programs to the rest of the cohort.

**Insight: The lowest-scoring students all lack test preparation.**

**Business Impact:** The weakest segment of the cohort is also the segment most likely to benefit most from intervention.

**Recommended Action:** Proactively identify and enroll no-preparation, low-scoring students in support programs.

**Insight: Mathematics is the cohort's strongest subject; Reading is the weakest.**

**Business Impact:** A one-size-fits-all support plan would under-serve literacy development.

**Recommended Action:** Design subject-specific support, with added focus on Reading and Writing.

**Insight: Subject scores are weakly correlated with each other.**

**Business Impact:** Strength in one subject does not predict strength in another, so students need individualized rather than blanket support plans.

**Recommended Action:** Use per-subject dashboards, not just the blended score, when assigning support resources.

**Insight: Half the cohort sits in the 60–70% score band with no student above 80%.**

**Business Impact:** There is a large, reachable “middle group” rather than a small at-risk tail, meaning broad-based interventions could shift many students upward.

**Recommended Action:** Pilot preparation or tutoring broadly rather than only for the lowest scorers.

**Insight: Ethnicity and gender differences are small (under 3 points).**

**Business Impact:** These fixed demographic traits are not useful levers for improving outcomes and should not be the basis for targeting interventions.

**Recommended Action:** Focus support programs on behavior (preparation, study habits) rather than demographic profiling.

**Insight: The parental-education relationship is non-linear and likely affected by small group sizes (13–23 students per group).**

**Business Impact:** Treating parental education as a reliable predictor risks drawing the wrong conclusion from limited data.

**Recommended Action:** Avoid using parental education alone to prioritize support; validate with a larger sample before acting on it.

**Insight: In the companion 300-student dataset, study hours nearly triple from Low to High performers (2.7 to 7.5).**

**Business Impact:** Study time is one of the clearest, most controllable levers associated with performance level.

**Recommended Action:** Encourage structured study-time goals as part of any academic support plan.

**Insight:** Assignment completion shows the widest gap between performance levels (260 vs 937).

**Business Impact:** Assignment completion rate may serve as an early-warning signal for at-risk students, ahead of final scores.

**Recommended Action:** Track assignment completion rates in real time to flag at-risk students before assessments.

## 16. Recommendations

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- Expand access to test preparation — it shows the largest, most consistent score gap (8–10 points) of any factor in the dataset, and 59% of students have not completed one.
- Use assignment-completion tracking as an early-warning system, given its strong association with performance level in the companion dataset.
- Build subject-specific support (particularly Reading and Writing) rather than a single blended “overall score” intervention, since subject scores are weakly correlated.
- Target support toward the 60–70% “middle band” (50 students) where the preparation-linked gains observed in Section 11 could plausibly lift the largest number of students.
- Proactively identify no-preparation, low-scoring students (Section 9) for direct outreach rather than waiting for results to decline further.
- Encourage structured study-time targets and screen-time awareness, reflecting the behavioral gradient seen across performance levels in the EDA dashboard.
- Avoid using ethnicity, gender, or parental education as primary targeting criteria for support programs — the data shows these are weak or unreliable differentiators compared to preparation and study habits.
- Continue monitoring these KPIs each term/cohort to confirm whether the test-preparation gap holds as more student data becomes available.

## 17. Conclusion

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Across both dashboards, the 100-student score dataset and the 300-student study-habit dataset tell a consistent story: academic performance in this data is driven far more by coachable behaviors — test preparation, study hours, assignment completion, and screen-time discipline — than by fixed demographic characteristics such as gender, ethnicity, or parental education, where differences were small or inconsistent.

The clearest opportunity is test preparation, associated with an 8–10 point improvement across every subject, currently accessed by less than half the core cohort. The clearest risk is the concentration of half the class in the 60–70% band with no student above 80%, which, without intervention, caps the cohort’s overall trajectory.

The two Power BI dashboards give academic stakeholders a fast, interactive way to monitor these metrics, filter to any student segment, and identify individual students who need support — turning a static spreadsheet of scores into a decision-support tool for targeted academic intervention.

## 18. Portfolio Project Summary

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**Project Name:** Student Performance Analytics Dashboard

**Tools Used:** Microsoft Excel, Power BI

**Business/Academic Objective:** Analyze a 100-student academic performance dataset and a complementary 300-student study-habit dataset to identify the demographic and behavioral factors most associated with strong or weak outcomes, and present the findings through interactive Power BI dashboards.

**Key Analysis:** Built KPI cards for subject and overall scores; compared performance across gender, ethnicity, and parental education; quantified the impact of test preparation; identified top and bottom performing students; and analyzed study hours, sleep, screen time, and assignment completion against a classified performance level.

**Key Findings:** Test preparation is the single strongest performance driver (8–10 point gap across all subjects), demographic factors show only small differences, and behavioral variables (study hours, screen time, assignment completion) scale strongly with performance level in the companion dataset.

**Business Value:** The dashboards let academic stakeholders move from static spreadsheets to interactive, filterable views that surface which students and factors need attention — supporting faster, evidence-based decisions on where to direct academic support resources.

## 19. Skills Demonstrated

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- Microsoft Excel (data organization and review)
- Data Cleaning & Data Preparation
- Data Analysis
- Exploratory Data Analysis (EDA)
- Power BI (dashboard development)
- Data Visualization
- KPI Development
- Statistical Comparison (group averages, correlation)
- Interactive Reporting (slicers/filters)
- Data Storytelling
- Insight Generation & Business/Academic Recommendations
- Data-Driven Decision Making